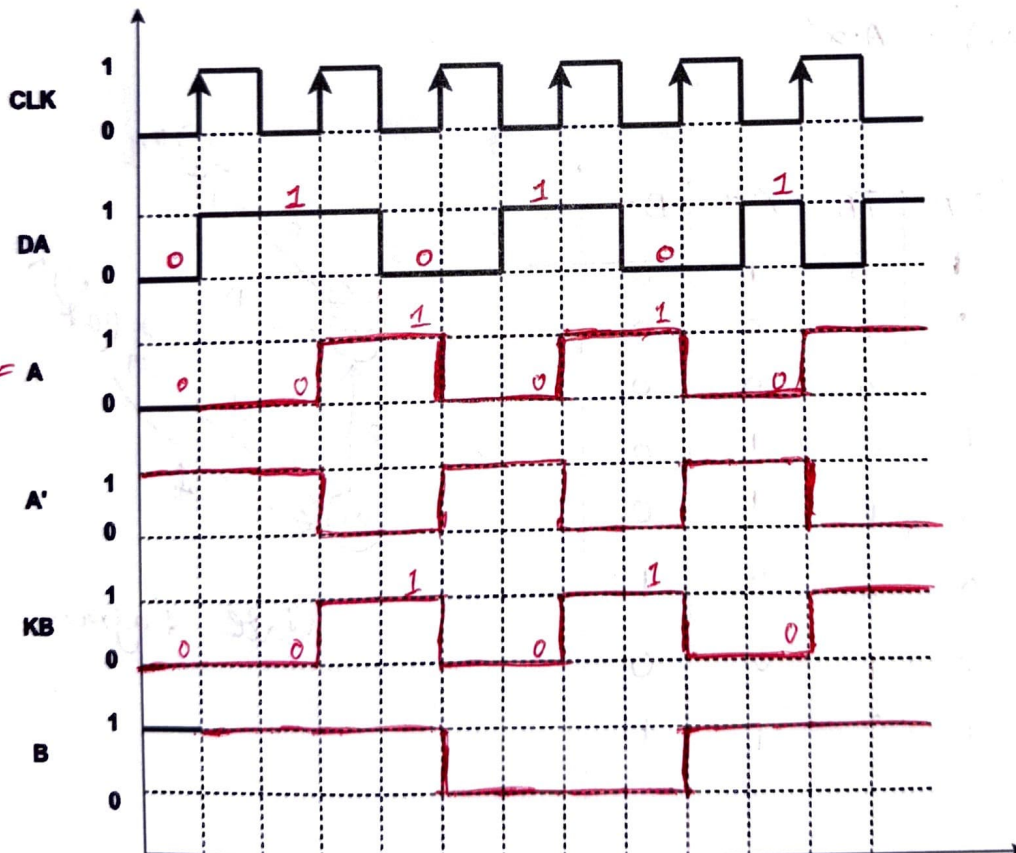
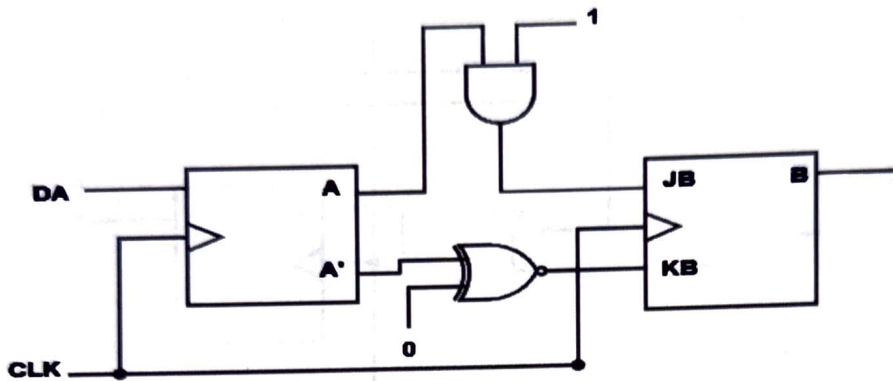


Name: *Solution*

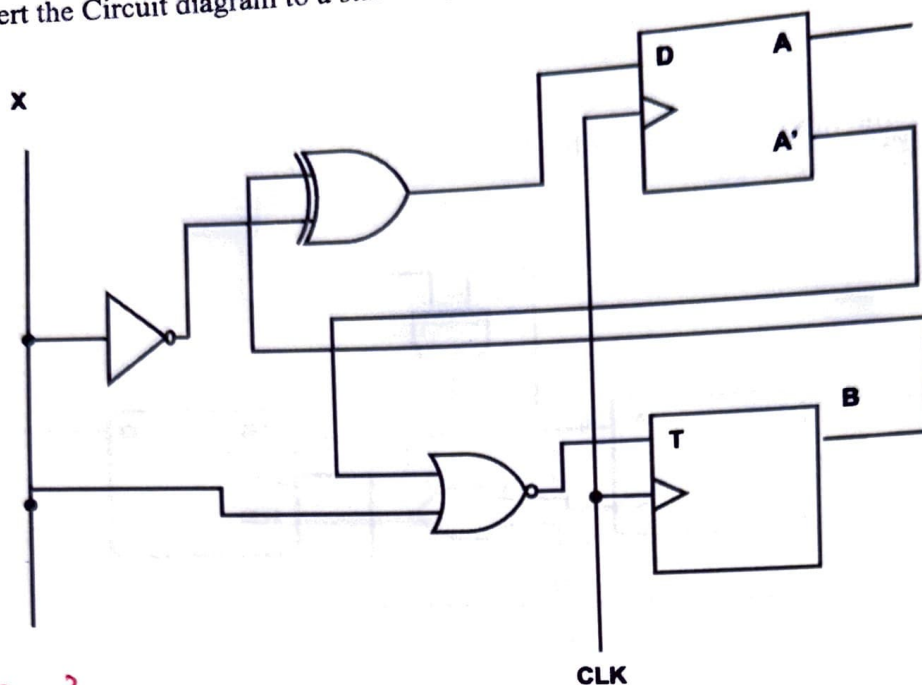
ID:

Section:

1. Complete the timing diagram for the following Flip-flop circuit diagram. The FF is a positive-edge-triggered.



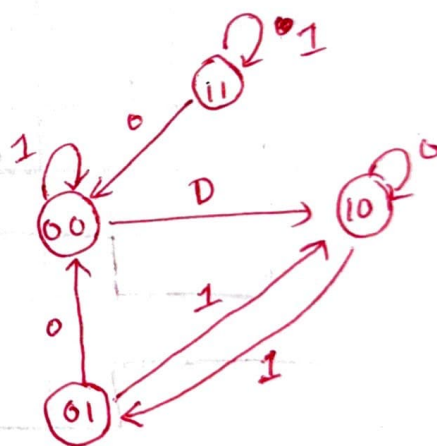
2. Convert the Circuit diagram to a state diagram.



$$DA = B \oplus X'$$

$$TB = (A' + X)' = A \cdot X'$$

A	B	X	DA	TB	A+	B+
0	0	0	1	0	1	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	1	0
1	0	0	1	1	1	0
1	0	1	0	0	0	1
1	1	0	0	1	0	0
1	1	1	1	0	1	1



State diagram